

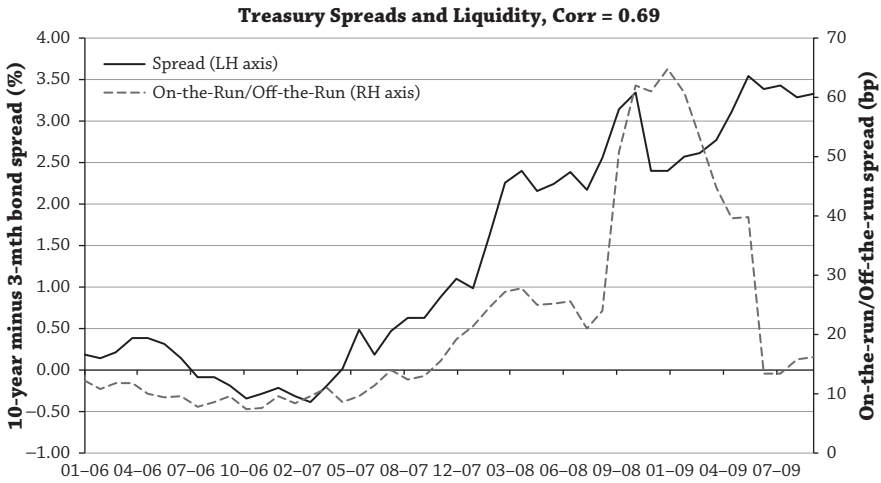


Figure 9.10

Treasuries are risk free) but different volumes and prices.²⁴ The effect is, however, influenced by how and when the Treasury brings to market new issues.

Figure 9.10 plots the ten-year Treasury minus three-month T-bill term spread (left-hand axis) with the on-the-run/off-the-run spread (on the right-hand axis) from January 2006 to October 2009. Higher on-the-run/off-the-run spreads indicate periods of greater illiquidity. During this sample, the correlation between the term spread and the illiquidity spread was a lofty 69%. Figure 9.10 shows that the term spread tracks the rise in illiquidity from 2007 to 2009. However, the term spread remains elevated after the economy stabilizes and the financial crisis passes post-2009, whereas the illiquidity spreads fall and the market recovers its liquidity. Nevertheless, the demand for Treasuries during times of illiquidity and high volatility, or the flight-to-safety effect, was an important factor driving Treasuries during the financial crisis.

Recently several researchers have focused on the effect of different investors, particularly foreign investors, on the dynamics of the yield curve. The top panel of Figure 9.11 graphs the proportion of Treasuries held by foreign investors (excluding official holdings in central banks), showing this has increased from below 10% during the 1960s to over 50% today. (The largest of these foreign positions is China's; see chapter 1.) The bottom panel of Figure 9.11 plots net new issues by the Treasury and foreign demand for Treasuries, both normalized by GDP. The panel shows that both are cyclical and the effects have been increasing over time. Foreign demand had a local peak in the mid-2000s and also during the financial crisis in 2008. Fed Chair Ben Bernanke himself suggested that these

²⁴ See Amihud and Mendelson (1991) and Krishnamurthy (2002).